Lesson 59: Empirical Risk Minimization and Generalization

AI/ML Mathematics

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1 Motivation and Overview

Consider training a neural network to classify images: we collect a dataset of n labeled examples $(x_1, y_1), \ldots, (x_n, y_n)$, choose a hypothesis class $\mathcal{H}$ (e.g., all neural networks of a given architecture), and select the function $\hat{f} \in \mathcal{H}$ that minimizes the average loss on the training data. This procedure—*empirical risk minimization* (ERM)—is the foundation of virtually every supervised learning algorithm. Yet training error alone is meaningless; what matters is the *population risk*, the expected loss on unseen data drawn from the same distribution. The central question of statistical learning theory is: *when and why does low training error imply low population risk?* Answering this question requires bounding the gap between empirical and population risk uniformly over the hypothesis class, leading to the classical notions of VC dimension and uniform convergence. The tools developed in this lesson will let us derive explicit generalization bounds that quantify how many training examples suffice to guarantee good out-of-sample performance.

Why ERM and Generalization Matter for RL/ML

- **Fitted value iteration**: At each step, a regression problem is solved by ERM over a hypothesis class; generalization bounds control the propagation of approximation error across Bellman backups.
- **Imitation learning (DAgger)**: The learner minimizes empirical classification loss against an expert's actions; generalization theory determines the sample complexity of achieving near-expert performance.
- **Neural network capacity**: VC dimension and uniform convergence provide the classical framework for understanding why overparameterized networks can still generalize, motivating the norm-based bounds of Lesson 61 and beyond.

Prerequisites. Probability spaces, random variables, expectation, and variance (Lesson 9–11). Hoeffding's inequality and the union bound (Lesson 12, Theorem 7.3). Convergence in probability (Lesson 12, Definition 5.1). Basic set theory and countability arguments (Lesson 1).

Learning objectives. After completing this lesson, the student will be able to:

- *Define* the population risk, empirical risk, hypothesis class, excess risk, and ERM, and *explain* the relationship between them.
- *Prove* the finite hypothesis class generalization bound using Hoeffding's inequality and the union bound.
- *Define* shattering, the growth function, and VC dimension, and *compute* the VC dimension of halfspaces in $\mathbb{R}^d$.
- *State* and *prove* Sauer's lemma bounding the growth function by the VC dimension.
- *State* the VC generalization bound and *explain* its dependence on VC dimension and sample size.
- *Prove* that ERM generalizes under uniform convergence and *state* the fundamental theorem of statistical learning for binary classification.

2 References

- Bach [2024, Chapters 1–2] – Statistical learning framework, ERM, finite class bounds, and introduction to VC theory.

- Shalev-Shwartz and Ben-David [2014, Chapters 2–6] – PAC learning, finite hypothesis classes, VC dimension, and the fundamental theorem of statistical learning.
- Mohri et al. [2018, Chapters 3–4] – Rademacher complexity and VC dimension with detailed proofs.
- Vapnik [1998, Chapters 6–7] – Original treatment of VC theory and structural risk minimization.
- Devroye et al. [1996, Chapter 12] – VC dimension, growth functions, and Sauer's lemma with combinatorial proofs.

3 Statistical Learning Framework

We begin by formalizing the supervised learning problem in its full generality.

Definition 3.1 (Supervised Learning Setup)**.** A *supervised learning problem* is specified by:

(i) An *input space* $\mathcal{X}$ (e.g., $\mathbb{R}^d$).
(ii) An *output space* $\mathcal{Y}$ (e.g., $\{0,1\}$ for binary classification, or $\mathbb{R}$ for regression).
(iii) An unknown *data-generating distribution* P on $\mathcal{X} \times \mathcal{Y}$.
(iv) A *loss function* $\ell\colon \mathcal{Y} \times \mathcal{Y} \to [0,\infty)$ measuring the cost of predicting $\hat{y}$ when the true label is y.
(v) A *hypothesis class* $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$, a set of functions $f\colon \mathcal{X} \to \mathcal{Y}$ from which the learner selects.

The learner observes a *training sample* $S = \{(X_1,Y_1),\dots,(X_n,Y_n)\}$ consisting of n independent draws from P, and must choose $\hat{f} \in \mathcal{H}$ based on S.

Definition 3.2 (Population Risk and Empirical Risk)**.** The *population risk* (or *true risk*) of a function $f\colon \mathcal{X} \to \mathcal{Y}$ is

$$R(f) = \mathbb{E}_{(X,Y)\sim P}\big[\ell(f(X),Y)\big].$$

The *empirical risk* (or *training error*) of f with respect to a sample $S = \{(X_i,Y_i)\}_{i=1}^n$ is

$$\hat{R}_n(f) = \frac{1}{n}\sum_{i=1}^n \ell(f(X_i),Y_i).$$

Remark 3.3. For a fixed f, the empirical risk $\hat{R}_n(f)$ is the sample mean of n i.i.d. random variables $Z_i = \ell(f(X_i),Y_i)$, each with expectation $R(f)$. By the law of large numbers (Lesson 12, Theorem 5.3), $\hat{R}_n(f) \to R(f)$ in probability as $n \to \infty$—but this holds *pointwise* for each fixed f, not uniformly over $\mathcal{H}$.

Definition 3.4 (Bayes Risk and Best-in-Class Risk)**.** The *Bayes-optimal predictor* is $f^* = \arg\min_{f\colon \mathcal{X}\to\mathcal{Y}} R(f)$, and $R(f^*)$ is the *Bayes risk*. The *best-in-class predictor* is $f^*_{\mathcal{H}} = \arg\min_{f\in\mathcal{H}} R(f)$, and $R(f^*_{\mathcal{H}})$ is the *best-in-class risk*.

Definition 3.5 (Empirical Risk Minimizer)**.** The *empirical risk minimizer* (ERM) over $\mathcal{H}$ is any function

$$\hat{f}_n = \arg\min_{f\in\mathcal{H}} \hat{R}_n(f).$$

When the minimum is not attained, $\hat{f}_n$ denotes any ε_n-approximate minimizer with $\varepsilon_n \to 0$.

Definition 3.6 (Excess Risk Decomposition)**.** The *excess risk* of the ERM $\hat{f}_n$ decomposes as

$$R(\hat{f}_n) - R(f^*) = \underbrace{\big(R(\hat{f}_n) - R(f^*_{\mathcal{H}})\big)}_{\text{estimation error}} + \underbrace{\big(R(f^*_{\mathcal{H}}) - R(f^*)\big)}_{\text{approximation error}}.$$

The *estimation error* measures how well ERM finds the best function within $\mathcal{H}$; the *approximation error* measures how well $\mathcal{H}$ approximates the Bayes-optimal predictor.

Example 3.7 (Linear Classifiers)**.** Let $\mathcal{X} = \mathbb{R}^d$, $\mathcal{Y} = \{-1, +1\}$, and $\ell(\hat{y}, y) = \mathbf{1}_{\{\hat{y} \neq y\}}$ (0-1 loss). The class of linear classifiers is $\mathcal{H} = \{x \mapsto \operatorname{sign}(w^\top x) : w \in \mathbb{R}^d\}$. The ERM selects the linear separator that minimizes the fraction of misclassified training points. The approximation error is zero if and only if the Bayes-optimal boundary is linear; otherwise, increasing the complexity of $\mathcal{H}$ (e.g., to polynomial classifiers) reduces approximation error at the cost of increased estimation error.

Example 3.8 (Polynomial Regression)**.** Let $\mathcal{X} = [0, 1]$, $\mathcal{Y} = \mathbb{R}$, and $\ell(\hat{y}, y) = (\hat{y} - y)^2$ (squared loss). The class of polynomials of degree at most k is $\mathcal{H}_k = \{x \mapsto \sum_{j=0}^{k} a_j x^j : a_0, \ldots, a_k \in \mathbb{R}\}$. ERM reduces to least-squares polynomial fitting. As k increases, the approximation error decreases (by the Weierstrass approximation theorem), but the estimation error grows because $\mathcal{H}_k$ becomes larger—a tension we will quantify precisely.

ERM in Reinforcement Learning

In fitted Q-iteration, the agent repeatedly solves regression problems of the form $\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} (f(s_i, a_i) - y_i)^2$, where $y_i = r_i + \gamma \max_{a'} \hat{Q}(s_i', a')$ are bootstrapped targets. Each regression step is an instance of ERM. The estimation error at each step propagates through the Bellman backup, and bounding the total error requires uniform convergence guarantees over $\mathcal{H}$ applied at every iteration.

We now ask: for a given hypothesis class $\mathcal{H}$, how large must n be to ensure that the estimation error of ERM is small? The simplest case—finite $\mathcal{H}$—already reveals the key ideas.

4 Finite Hypothesis Class Bounds

When $\mathcal{H}$ is finite, we can apply Hoeffding's inequality to each $f \in \mathcal{H}$ individually and then take a union bound. This yields the fundamental generalization bound for finite classes.

Assumption 4.1 (Bounded Loss)**.** Throughout this section, we assume the loss is bounded: $0 \leq \ell(f(x), y) \leq B$ for all $f \in \mathcal{H}$, $x \in \mathcal{X}$, and $y \in \mathcal{Y}$.

Lemma 4.2 (Pointwise Concentration)**.** *Under Assumption 4.1, for any fixed $f \in \mathcal{H}$ and any $\varepsilon > 0$,*

$$\mathbb{P}\big(|\hat{R}_n(f) - R(f)| \geq \varepsilon\big) \leq 2 \exp\left(-\frac{2n\varepsilon^2}{B^2}\right).$$

Proof. The random variables $Z_i = \ell(f(X_i), Y_i)$ are i.i.d. with $\mathbb{E}[Z_i] = R(f)$ and $0 \leq Z_i \leq B$. Since $\hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^{n} Z_i$, the result follows directly from Hoeffding's inequality for bounded i.i.d. random variables (Lesson 12, Corollary 7.5). □

Theorem 4.3 (Finite Class Generalization Bound)**.** *Let $\mathcal{H}$ be a finite hypothesis class with $|\mathcal{H}| < \infty$. Under Assumption 4.1, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the draw of the training sample S,*

$$\sup_{f \in \mathcal{H}} |\hat{R}_n(f) - R(f)| \leq B \sqrt{\frac{\ln(2|\mathcal{H}|/\delta)}{2n}}.$$

In particular, for the ERM $\hat{f}_n$,

$$R(\hat{f}_n) - \min_{f \in \mathcal{H}} R(f) \leq 2B \sqrt{\frac{\ln(2|\mathcal{H}|/\delta)}{2n}}.$$

Proof. Set $\varepsilon = B\sqrt{\ln(2|\mathcal{H}|/\delta)/(2n)}$. By Lemma 4.2, for each fixed $f \in \mathcal{H}$,

$$\mathbb{P}\big(|\hat{R}_n(f) - R(f)| \geq \varepsilon\big) \leq 2\exp\left(-\frac{2n\varepsilon^2}{B^2}\right) = 2\exp\left(-\ln\frac{2|\mathcal{H}|}{\delta}\right) = \frac{\delta}{|\mathcal{H}|}.$$

By the union bound over all $f \in \mathcal{H}$,

$$\begin{aligned}\mathbb{P}\left(\sup_{f\in\mathcal{H}} |\hat{R}_n(f) - R(f)| \geq \varepsilon\right) &\leq \sum_{f\in\mathcal{H}} \mathbb{P}\big(|\hat{R}_n(f) - R(f)| \geq \varepsilon\big) \\ &\leq |\mathcal{H}| \cdot \frac{\delta}{|\mathcal{H}|} = \delta.\end{aligned}$$

Thus with probability at least $1-\delta$, $|\hat{R}_n(f) - R(f)| \leq \varepsilon$ for all $f \in \mathcal{H}$ simultaneously.

For the excess risk bound, let $f^*_{\mathcal{H}} = \arg\min_{f\in\mathcal{H}} R(f)$. On the event that uniform convergence holds:

$$\begin{aligned}R(\hat{f}_n) - R(f^*_{\mathcal{H}}) &= \big(R(\hat{f}_n) - \hat{R}_n(\hat{f}_n)\big) + \big(\hat{R}_n(\hat{f}_n) - \hat{R}_n(f^*_{\mathcal{H}})\big) + \big(\hat{R}_n(f^*_{\mathcal{H}}) - R(f^*_{\mathcal{H}})\big) \\ &\leq \varepsilon + 0 + \varepsilon = 2\varepsilon,\end{aligned}$$

where we used that $\hat{f}_n$ minimizes $\hat{R}_n$ so $\hat{R}_n(\hat{f}_n) \leq \hat{R}_n(f^*_{\mathcal{H}})$. □

Example 4.4 (Threshold Classifiers)**.** Let $\mathcal{X} = [0,1]$, $\mathcal{Y} = \{0,1\}$, and consider the finite set of threshold classifiers at multiples of $1/N$: $\mathcal{H}_N = \{f_j : j = 0,1,\ldots,N\}$ where $f_j(x) = \mathbf{1}_{\{x>j/N\}}$. Then $|\mathcal{H}_N| = N+1$. With 0-1 loss ($B=1$), Theorem 4.3 gives: with probability at least $1-\delta$,

$$R(\hat{f}_n) - \min_{f\in\mathcal{H}_N} R(f) \leq 2\sqrt{\frac{\ln(2(N+1)/\delta)}{2n}}.$$

For $N = 1000$ and $\delta = 0.05$, achieving excess risk at most 0.1 requires $n \geq 200\ln(40002) \approx 2122$ samples.

Finite Class Generalization Bound

For a finite hypothesis class $\mathcal{H}$ with bounded loss $\ell \in [0,B]$, the estimation error of ERM satisfies

$$R(\hat{f}_n) - \min_{f\in\mathcal{H}} R(f) \leq O\left(B\sqrt{\frac{\log|\mathcal{H}|}{n}}\right)$$

with high probability. The sample complexity to achieve excess risk $\leq \varepsilon$ is $n = O\left(\frac{B^2\log|\mathcal{H}|}{\varepsilon^2}\right)$. The dependence on $\log|\mathcal{H}|$ (not $|\mathcal{H}|$) is crucial: it means the sample complexity grows only logarithmically with the number of hypotheses.

The Union Bound is Loose for Continuous Classes

The union bound argument fails when $\mathcal{H}$ is infinite (or even uncountable), since we cannot sum over all hypotheses. Naively, one might expect that infinite hypothesis classes require infinitely many samples. This is false: the relevant quantity is not the cardinality of $\mathcal{H}$ but its *effective complexity*, measured by the VC dimension or Rademacher complexity.

The finite class bound motivates a fundamental question: can we extend the union bound argument to infinite hypothesis classes by replacing $|\mathcal{H}|$ with a measure of effective complexity? The answer is yes, through the notion of VC dimension.

5 VC Dimension

We restrict attention to binary classification: $\mathcal{Y} = \{0, 1\}$ and $\ell(\hat{y}, y) = \mathbf{1}_{\{\hat{y} \neq y\}}$ (0-1 loss). The key insight is that even though $\mathcal{H}$ may be uncountably infinite, it can only induce finitely many distinct labelings on any finite set of points.

Definition 5.1 (Restriction and Shattering)**.** Let $\mathcal{H} \subseteq \{0, 1\}^{\mathcal{X}}$ be a hypothesis class and $C = \{x_1, \ldots, x_m\} \subseteq \mathcal{X}$ a finite set of m points.

(i) The *restriction* of $\mathcal{H}$ to C is $\mathcal{H}|_C = \{(f(x_1), \ldots, f(x_m)) : f \in \mathcal{H}\} \subseteq \{0, 1\}^m$.

(ii) The class $\mathcal{H}$ *shatters* C if $\mathcal{H}|_C = \{0, 1\}^m$, i.e., every possible labeling of C is realized by some $f \in \mathcal{H}$.

Definition 5.2 (Growth Function)**.** The *growth function* (or *shattering coefficient*) of $\mathcal{H}$ is

$$\Pi_{\mathcal{H}}(m) = \max_{C \subseteq \mathcal{X},\, |C| = m} |\mathcal{H}|_C|.$$

It counts the maximum number of distinct labelings that $\mathcal{H}$ can produce on any set of m points. We always have $\Pi_{\mathcal{H}}(m) \leq 2^m$.

Definition 5.3 (VC Dimension)**.** The *Vapnik-Chervonenkis (VC) dimension* of $\mathcal{H}$, denoted $\mathrm{VC}(\mathcal{H})$, is the largest integer d such that there exists a set of d points that is shattered by $\mathcal{H}$:

$$\mathrm{VC}(\mathcal{H}) = \max\{d \in \mathbb{N} : \Pi_{\mathcal{H}}(d) = 2^d\}.$$

If $\mathcal{H}$ shatters arbitrarily large sets, then $\mathrm{VC}(\mathcal{H}) = \infty$.

Remark 5.4. To show $\mathrm{VC}(\mathcal{H}) \geq d$, it suffices to exhibit *one* set of d points that is shattered. To show $\mathrm{VC}(\mathcal{H}) < d$, one must prove that *no* set of d points is shattered.

Example 5.5 (VC Dimension of Intervals)**.** Let $\mathcal{X} = \mathbb{R}$ and $\mathcal{H} = \{x \mapsto \mathbf{1}_{\{a \leq x \leq b\}} : a \leq b\}$ (interval classifiers). We show $\mathrm{VC}(\mathcal{H}) = 2$.

Lower bound: The two points $\{1, 2\}$ are shattered. The four labelings $(0, 0), (1, 0), (0, 1), (1, 1)$ are realized by the intervals $(-\infty, 0]$, $[0.5, 1.5]$, $[1.5, 2.5]$, and $[0.5, 2.5]$, respectively.

Upper bound: Consider any three points $x_1 < x_2 < x_3$. The labeling $(1, 0, 1)$ cannot be realized by any interval: if x_1 and x_3 are inside $[a, b]$, then $x_2 \in [x_1, x_3] \subseteq [a, b]$, so $f(x_2) = 1 \neq 0$. Hence no set of three points is shattered, so $\mathrm{VC}(\mathcal{H}) \leq 2$.

Example 5.6 (VC Dimension of Halfspaces)**.** Let $\mathcal{X} = \mathbb{R}^d$ and $\mathcal{H} = \{x \mapsto \mathbf{1}_{\{w^\top x + b \geq 0\}} : w \in \mathbb{R}^d,\, b \in \mathbb{R}\}$ (halfspace classifiers). We prove $\mathrm{VC}(\mathcal{H}) = d + 1$.

Lower bound ($\mathrm{VC} \geq d + 1$*)*: Consider the $d + 1$ points $\{0, e_1, e_2, \ldots, e_d\}$ where e_i is the i-th standard basis vector. We claim $\mathcal{H}$ shatters this set. For any labeling $(y_0, y_1, \ldots, y_d) \in \{0, 1\}^{d+1}$, define $\tilde{y}_i = 2y_i - 1 \in \{-1, +1\}$. We need $w \in \mathbb{R}^d, b \in \mathbb{R}$ with $\mathrm{sign}(b) = \tilde{y}_0$ and $\mathrm{sign}(w_i + b) = \tilde{y}_i$ for $i = 1, \ldots, d$. Setting $b = \tilde{y}_0$ and $w_i = \tilde{y}_i - b$ for each i achieves this, since $w_i + b = \tilde{y}_i$ and $b = \tilde{y}_0$. (More precisely, we use the convention that $\mathrm{sign}(0) = 1$ and scale w, b by a small factor to avoid ties.)

Upper bound ($\mathrm{VC} \leq d + 1$*)*: By Radon's theorem, any set of $d + 2$ points in $\mathbb{R}^d$ can be partitioned into two disjoint sets A, B such that $\mathrm{conv}(A) \cap \mathrm{conv}(B) \neq \emptyset$. No hyperplane can separate A from B, so the labeling that assigns 1 to A and 0 to B is not realizable. Hence no set of $d + 2$ points is shattered.

Example 5.7 (VC Dimension of Finite Classes)**.** If $|\mathcal{H}| < \infty$, then $\Pi_{\mathcal{H}}(m) \leq |\mathcal{H}|$ for all m. Shattering a set of d points requires $2^d \leq |\mathcal{H}|$, so $\mathrm{VC}(\mathcal{H}) \leq \log_2 |\mathcal{H}|$.

We now establish the key combinatorial result that limits the growth function once m exceeds the VC dimension.

5.1 Sauer's Lemma

Theorem 5.8 (Sauer's Lemma)**.** *Let $\mathcal{H}$ be a hypothesis class with* $\mathrm{VC}(\mathcal{H}) = d < \infty$*. Then for all $m \geq 1$,*

$$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^{d} \binom{m}{i}.$$

In particular, for $m \geq d$,

$$\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d.$$

Proof. We prove the first inequality by induction on $m + d$.

Base cases: If $d = 0$, then $\mathcal{H}$ shatters no single point, so $\mathcal{H}|_C$ contains at most one labeling for every C; thus $\Pi_{\mathcal{H}}(m) \leq 1 = \binom{m}{0}$. If $m = 0$, then $\Pi_{\mathcal{H}}(0) = 1 \leq \sum_{i=0}^{d} \binom{0}{i} = 1$.

Inductive step: Fix $m \geq 1$ and $d \geq 1$, and assume the result holds for all pairs (m', d') with $m' + d' < m + d$. Take any set $C = \{x_1, \ldots, x_m\}$ achieving the maximum in the growth function. Partition $\mathcal{H}|_C$ into pairs based on the last coordinate:

$$\begin{aligned}
\mathcal{H}_0 &= \{(b_1, \ldots, b_{m-1}) : (b_1, \ldots, b_{m-1}, 0) \in \mathcal{H}|_C \text{ or } (b_1, \ldots, b_{m-1}, 1) \in \mathcal{H}|_C\}, \\
\mathcal{H}_1 &= \{(b_1, \ldots, b_{m-1}) : (b_1, \ldots, b_{m-1}, 0) \in \mathcal{H}|_C \text{ and } (b_1, \ldots, b_{m-1}, 1) \in \mathcal{H}|_C\}.
\end{aligned}$$

Then $|\mathcal{H}|_C| = |\mathcal{H}_0| + |\mathcal{H}_1|$. Now $\mathcal{H}_0$ is the restriction of $\mathcal{H}$ to $C' = \{x_1, \ldots, x_{m-1}\}$, so $|\mathcal{H}_0| \leq \Pi_{\mathcal{H}}(m-1)$. The set $\mathcal{H}_1$ records the labelings on C' for which both labels of x_m are achievable; $\mathcal{H}_1$ can be viewed as the restriction of some class $\mathcal{H}'$ with $\mathrm{VC}(\mathcal{H}') \leq d - 1$ (because if $\mathcal{H}_1$ shattered a set $D \subseteq C'$ of size d, then $\mathcal{H}$ would shatter $D \cup \{x_m\}$, a set of size $d + 1$, contradicting $\mathrm{VC}(\mathcal{H}) = d$). Hence $|\mathcal{H}_1| \leq \Pi_{\mathcal{H}'}(m - 1)$.

By the inductive hypothesis:

$$\begin{aligned}
|\mathcal{H}|_C| &\leq \Pi_{\mathcal{H}}(m-1) + \Pi_{\mathcal{H}'}(m-1) \\
&\leq \sum_{i=0}^{d} \binom{m-1}{i} + \sum_{i=0}^{d-1} \binom{m-1}{i} \\
&= \sum_{i=0}^{d} \binom{m-1}{i} + \sum_{i=1}^{d} \binom{m-1}{i-1} = \sum_{i=0}^{d} \left[\binom{m-1}{i} + \binom{m-1}{i-1}\right],
\end{aligned}$$

where we set $\binom{m-1}{-1} = 0$. By Pascal's identity $\binom{m-1}{i} + \binom{m-1}{i-1} = \binom{m}{i}$, this gives $|\mathcal{H}|_C| \leq \sum_{i=0}^{d} \binom{m}{i}$.

For the second inequality, when $m \geq d$:

$$\sum_{i=0}^{d} \binom{m}{i} \leq \sum_{i=0}^{d} \binom{m}{i} \left(\frac{m}{d}\right)^{d-i} \leq \left(\frac{m}{d}\right)^d \sum_{i=0}^{d} \binom{m}{i} \frac{d^i}{m^i}.$$

A cleaner argument uses the bound $\sum_{i=0}^{d} \binom{m}{i} \leq (em/d)^d$, which follows from the binomial theorem: $(1+1)^m = \sum_{i=0}^{m} \binom{m}{i} \geq \binom{m}{d} \geq (m/d)^d$, so $\sum_{i=0}^{d} \binom{m}{i} \leq (m/d)^d \cdot (e)^d = (em/d)^d$ by the inequality $\sum_{i=0}^{d} \binom{m}{i} / \binom{m}{d} \leq e^d$ (which holds because $\binom{m}{i} / \binom{m}{d} \leq (d/m)^{d-i}$ for $i \leq d$ and the geometric series sums to at most e^d when $d \leq m$; see Devroye et al. [1996, Lemma 12.3] for a detailed proof). □

Remark 5.9. Sauer's lemma is the bridge between combinatorics and learning theory. Before the VC dimension is reached ($m \leq d$), the growth function can be 2^m (exponential). Once $m > d$, the growth function is bounded by a polynomial of degree d in m. This polynomial growth is what makes learning from infinite hypothesis classes possible.

Sauer's Lemma and Polynomial Growth

If $\mathrm{VC}(\mathcal{H}) = d < \infty$, then the growth function satisfies

$$\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d \quad \text{for all } m \geq d.$$

The growth function transitions from potentially exponential (2^m) for $m \leq d$ to polynomial ($O(m^d)$) for $m > d$. This transition at $m = d$ is the essence of why VC dimension controls generalization.

With Sauer's lemma in hand, we can state the VC generalization bound.

5.2 The VC Generalization Bound

Theorem 5.10 (VC Bound)**.** *Let $\mathcal{H}$ be a class of binary classifiers with $\mathrm{VC}(\mathcal{H}) = d < \infty$. Then for any distribution P and any $\delta \in (0,1)$, with probability at least $1-\delta$ over a sample of size n,*

$$\sup_{f \in \mathcal{H}} |\hat{R}_n(f) - R(f)| \leq \sqrt{\frac{8}{n}\left(d \ln \frac{2en}{d} + \ln \frac{4}{\delta}\right)}.$$

Equivalently, for the ERM $\hat{f}_n$,

$$R(\hat{f}_n) \leq R(f^*_{\mathcal{H}}) + O\left(\sqrt{\frac{d \log(n/d)}{n}}\right).$$

The full proof of Theorem 5.10 uses a symmetrization argument (introducing "ghost samples") combined with Sauer's lemma. We sketch the key steps.

Proof sketch. The proof proceeds in three stages:

Step 1 (Symmetrization): Introduce an independent "ghost sample" $S' = \{(X'_1, Y'_1), \ldots, (X'_n, Y'_n)\}$ drawn from P. A standard symmetrization argument (see Shalev-Shwartz and Ben-David [2014, Chapter 4]) shows

$$\mathbb{P}\left(\sup_{f \in \mathcal{H}} |\hat{R}_n(f) - R(f)| \geq \varepsilon\right) \leq 2\mathbb{P}\left(\sup_{f \in \mathcal{H}} |\hat{R}_n(f) - \hat{R}'_n(f)| \geq \varepsilon/2\right),$$

where $\hat{R}'_n(f) = \frac{1}{n}\sum_{i=1}^n \ell(f(X'_i), Y'_i)$ is the empirical risk on the ghost sample.

Step 2 (Reduction to finite class): Conditioned on the combined sample $S \cup S'$ of $2n$ points, the supremum over $\mathcal{H}$ depends only on the restriction $\mathcal{H}|_{S \cup S'}$, which has at most $\Pi_{\mathcal{H}}(2n)$ elements. We apply the finite class bound (Theorem 4.3) to this restricted class.

Step 3 (Apply Sauer's lemma): By Sauer's lemma (Theorem 5.8), $\Pi_{\mathcal{H}}(2n) \leq (2en/d)^d$. Substituting into the finite class bound and solving for ε yields the stated result. □

Example 5.11 (VC Bound for Halfspaces)**.** For halfspace classifiers in $\mathbb{R}^d$ with $\mathrm{VC}(\mathcal{H}) = d+1$ (Example 5.6), the VC bound gives: with high probability,

$$R(\hat{f}_n) - R(f^*_{\mathcal{H}}) = O\left(\sqrt{\frac{(d+1)\log(n/(d+1))}{n}}\right).$$

To achieve excess risk ε, the sample complexity is $n = O(d/\varepsilon^2)$ (ignoring logarithmic factors). This matches the intuition that learning a d-dimensional linear classifier requires $O(d)$ samples per unit of accuracy.

With the VC bound established, we now formalize the key concept that connects all of these results: uniform convergence.

6 Uniform Convergence

Definition 6.1 (Uniform Convergence of Empirical Risk)**.** A hypothesis class $\mathcal{H}$ has the *uniform convergence property* if for every $\varepsilon > 0$ and every distribution P,

$$\lim_{n\to\infty} \mathbb{P}\left(\sup_{f\in\mathcal{H}} |\hat{R}_n(f) - R(f)| \geq \varepsilon\right) = 0.$$

If the convergence holds at an explicit rate, we say $\mathcal{H}$ has *uniform convergence at rate* $r(n,\delta)$: with probability at least $1-\delta$, $\sup_{f\in\mathcal{H}} |\hat{R}_n(f) - R(f)| \leq r(n,\delta)$.

Theorem 6.2 (ERM Generalizes Under Uniform Convergence)**.** *If $\mathcal{H}$ has uniform convergence at rate $r(n,\delta)$, then the ERM $\hat{f}_n$ satisfies: with probability at least $1-\delta$,*

$$R(\hat{f}_n) \leq \min_{f\in\mathcal{H}} R(f) + 2r(n,\delta).$$

Proof. Let $f^*_{\mathcal{H}} = \arg\min_{f\in\mathcal{H}} R(f)$. On the event $\{\sup_{f\in\mathcal{H}} |\hat{R}_n(f) - R(f)| \leq r(n,\delta)\}$ (which occurs with probability at least $1-\delta$), we have:

$$\begin{aligned}
R(\hat{f}_n) &\leq \hat{R}_n(\hat{f}_n) + r(n,\delta) && \text{(uniform convergence applied to } \hat{f}_n\text{)}\\
&\leq \hat{R}_n(f^*_{\mathcal{H}}) + r(n,\delta) && \text{(definition of ERM: } \hat{f}_n \text{ minimizes } \hat{R}_n\text{)}\\
&\leq R(f^*_{\mathcal{H}}) + r(n,\delta) + r(n,\delta) && \text{(uniform convergence applied to } f^*_{\mathcal{H}}\text{)}\\
&= R(f^*_{\mathcal{H}}) + 2r(n,\delta). && \square
\end{aligned}$$

Remark 6.3. Theorem 6.2 reveals the structure of ERM generalization: the factor of 2 arises because uniform convergence is invoked twice — once to convert from population to empirical risk for $\hat{f}_n$, and once in the opposite direction for $f^*_{\mathcal{H}}$. This is the fundamental proof template for all ERM generalization results.

The following definition captures the standard notion of learnability.

Definition 6.4 (PAC Learnability)**.** A hypothesis class $\mathcal{H}$ is *PAC learnable* (Probably Approximately Correct) if there exists an algorithm A and a function $n_{\mathcal{H}}\colon (0,1)^2 \to \mathbb{N}$ such that for every $\varepsilon, \delta \in (0,1)$ and every distribution P, when given a sample S of size $n \geq n_{\mathcal{H}}(\varepsilon,\delta)$, the algorithm outputs $\hat{f} = A(S)$ satisfying

$$\mathbb{P}\big(R(\hat{f}) \leq \min_{f\in\mathcal{H}} R(f) + \varepsilon\big) \geq 1-\delta.$$

The function $n_{\mathcal{H}}(\varepsilon,\delta)$ is the *sample complexity* of learning $\mathcal{H}$.

We can now state the crowning result of VC theory.

Theorem 6.5 (Fundamental Theorem of Statistical Learning)**.** *For binary classification with 0-1 loss, the following are equivalent for a hypothesis class $\mathcal{H}$:*

(i) $\mathcal{H}$ has the uniform convergence property.
(ii) ERM is a successful PAC learner for $\mathcal{H}$.
(iii) $\mathcal{H}$ is PAC learnable (by some algorithm).
(iv) $\mathrm{VC}(\mathcal{H}) < \infty$.

Moreover, the sample complexity of PAC learning $\mathcal{H}$ satisfies

$$C_1 \frac{d + \ln(1/\delta)}{\varepsilon^2} \leq n_{\mathcal{H}}(\varepsilon,\delta) \leq C_2 \frac{d\ln(1/\varepsilon) + \ln(1/\delta)}{\varepsilon^2},$$

where $d = \mathrm{VC}(\mathcal{H})$ and C_1, C_2 are universal constants.

Proof. We prove the key implications, deferring the full proof of $(iii) \Rightarrow (iv)$ to the references.

$(iv) \Rightarrow (i)$: This is the content of the VC bound (Theorem 5.10). If $\mathrm{VC}(\mathcal{H}) = d < \infty$, then $\sup_{f \in \mathcal{H}} |\hat{R}_n(f) - R(f)| \leq O(\sqrt{d \log(n/d)/n})$, which tends to 0 as $n \to \infty$.

$(i) \Rightarrow (ii)$: This is Theorem 6.2. Uniform convergence at rate $r(n, \delta)$ implies ERM achieves excess risk at most $2r(n, \delta)$.

$(ii) \Rightarrow (iii)$: Trivial, since ERM is a particular algorithm.

$(iii) \Rightarrow (iv)$: This is the converse direction and is the most subtle. The idea is that if $\mathrm{VC}(\mathcal{H}) = \infty$, then for any sample size n, there exists a distribution P under which no algorithm can PAC-learn $\mathcal{H}$ (a "no free lunch" argument; see Shalev-Shwartz and Ben-David [2014, Theorem 5.1]). □

Fundamental Theorem of Statistical Learning

For binary classification, a hypothesis class $\mathcal{H}$ is learnable if and only if $\mathrm{VC}(\mathcal{H}) < \infty$. Moreover, ERM is a universal learning algorithm: it succeeds for any learnable class. The sample complexity is $\Theta(d/\varepsilon^2)$ where $d = \mathrm{VC}(\mathcal{H})$, with the VC dimension serving as the unique measure of learning complexity for binary classification.

VC Theory and RL Function Approximation

In reinforcement learning, the value function V^π or Q^π is typically approximated by a class $\mathcal{H}$ (e.g., linear functions, neural networks). The VC dimension of $\mathcal{H}$ controls the sample complexity of learning a good approximation from observed transitions. For example, the PAC-MDP framework guarantees that an agent can learn a near-optimal policy with a number of samples polynomial in $\mathrm{VC}(\mathcal{H})$ and the MDP parameters. However, for deep networks, the classical VC dimension is often vacuous (too large), motivating the norm-based complexity measures developed in Lesson 61.

7 Exercises

Exercise 7.1 (VC Dimension of Rectangles). $(*)$
Let $\mathcal{X} = \mathbb{R}^2$ and $\mathcal{H} = \{(x_1, x_2) \mapsto \mathbf{1}_{\{a_1 \leq x_1 \leq b_1,\, a_2 \leq x_2 \leq b_2\}} : a_1 \leq b_1,\, a_2 \leq b_2\}$ (axis-aligned rectangles). Show that $\mathrm{VC}(\mathcal{H}) = 4$ by exhibiting a shattered set of 4 points and proving that no set of 5 points is shattered.

Exercise 7.2 (Generalization Bound Computation). $(*)$
A binary classifier uses halfspaces in $\mathbb{R}^{10}$ (so $\mathrm{VC}(\mathcal{H}) = 11$) and is trained on $n = 1000$ examples with 0-1 loss.

(a) Compute the upper bound on the estimation error from Theorem 5.10 with $\delta = 0.05$.
(b) How many samples would be needed to reduce this bound to 0.05?

Exercise 7.3 (Union of Hypothesis Classes). $(**)$
Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be hypothesis classes with $\mathrm{VC}(\mathcal{H}_1) = d_1$ and $\mathrm{VC}(\mathcal{H}_2) = d_2$.

(a) Show that $\Pi_{\mathcal{H}_1 \cup \mathcal{H}_2}(m) \leq \Pi_{\mathcal{H}_1}(m) + \Pi_{\mathcal{H}_2}(m)$.
(b) Prove that $\mathrm{VC}(\mathcal{H}_1 \cup \mathcal{H}_2) \leq 2(d_1 + d_2 + 1)\ln(2e)$.
Hint: Use Sauer's lemma and the bound $(em/d)^d + (em/d')^{d'} < 2^m$.

Exercise 7.4 (ERM for Fitted Value Iteration). $(**)$
In fitted value iteration, at each step k the agent solves $\hat{f}_k = \arg\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n (f(s_i) - T\hat{f}_{k-1}(s_i))^2$, where T is the Bellman operator and $s_1, \ldots, s_n$ are sampled states. Assume the loss $\ell(f(s), T\hat{f}_{k-1}(s)) = (f(s) - T\hat{f}_{k-1}(s))^2 \in [0, B^2]$ is bounded.

(a) Apply Theorem 4.3 (or its extension via VC theory) to bound the single-step estimation error $R(\hat{f}_k) - \min_{f \in \mathcal{H}} R(f)$.
(b) If K Bellman backups are performed and each uses an independent sample of size n, how does the total approximation error grow with K?

Exercise 7.5 (Growth Function Computation)**.** $(***)$
Let $\mathcal{X} = \mathbb{R}$ and $\mathcal{H}_k = \{x \mapsto \mathbf{1}_{\{p(x) \geq 0\}} : p \text{ is a polynomial of degree} \leq k\}$.
(a) Show that $\mathrm{VC}(\mathcal{H}_k) = k + 1$.
Hint: For the lower bound, choose $k + 1$ distinct points and use Lagrange interpolation. For the upper bound, note that a degree-k polynomial has at most k real roots.
(b) Compute $\Pi_{\mathcal{H}_2}(m)$ exactly for $m = 1, 2, 3, 4$ and verify that Sauer's lemma holds in each case.

8 Summary

- The *statistical learning framework* formalizes supervised learning via a data-generating distribution P, a loss function ℓ, and a hypothesis class $\mathcal{H}$; the goal is to minimize the population risk $R(f) = \mathbb{E}[\ell(f(X), Y)]$.
- *Empirical risk minimization* (ERM) selects $\hat{f}_n = \arg\min_{f \in \mathcal{H}} \hat{R}_n(f)$; the excess risk decomposes into approximation error (bias) and estimation error (variance).
- For *finite hypothesis classes*, the generalization bound $R(\hat{f}_n) - R(f^*_{\mathcal{H}}) \leq O(B\sqrt{\log |\mathcal{H}|/n})$ follows from Hoeffding's inequality and the union bound.
- The *VC dimension* $d = \mathrm{VC}(\mathcal{H})$ is the largest number of points that $\mathcal{H}$ can shatter; for halfspaces in $\mathbb{R}^d$, $\mathrm{VC} = d + 1$.
- *Sauer's lemma* shows that the growth function satisfies $\Pi_{\mathcal{H}}(m) \leq (em/d)^d$ for $m \geq d$, transitioning from exponential to polynomial growth at the VC dimension.
- The *VC generalization bound* gives $R(\hat{f}_n) - R(f^*_{\mathcal{H}}) = O(\sqrt{d \log(n/d)/n})$, with sample complexity $\Theta(d/\varepsilon^2)$.
- *Uniform convergence* of empirical to population risk is both necessary and sufficient for ERM to generalize; this is the mechanism underlying all classical generalization bounds.
- The *fundamental theorem of statistical learning* establishes the equivalence: $\mathcal{H}$ is PAC learnable if and only if $\mathrm{VC}(\mathcal{H}) < \infty$, with ERM as a universal learner.

9 Further Reading

- **Lesson 60** develops the bias-variance decomposition and model selection, providing the tools to choose the optimal hypothesis class complexity from the nested family $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \cdots$.
- **Lesson 61** (Approximation Theory) extends these ideas to neural networks, where the classical VC dimension is too large to be useful; Rademacher complexity and norm-based bounds provide tighter generalization guarantees.
- Vapnik and Chervonenkis [1971] is the original paper introducing VC dimension; Sauer [1972] and Shelah independently proved the lemma bounding growth functions.
- Valiant [1984] introduced the PAC learning framework, initiating the study of computational learning theory.

- Boucheron et al. [2013] provides a modern treatment of concentration inequalities and their applications to learning theory, going well beyond Hoeffding's inequality.
- For connections to RL, the PAC-MDP framework of Strehl, Li, and Littman (2009) uses VC-type arguments to bound the sample complexity of model-based reinforcement learning.

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